Project Brief: UrbanFix (Group C)

1. Problem Statement & Target User Group

- **Target User Group:** Active urban citizens and daily commuters (drivers, cyclists, and pedestrians) who navigate city infrastructure regularly and care about safety and public maintenance.
- **The Task:** Quickly and effortlessly reporting infrastructural defects (such as potholes, malfunctioning streetlights, and damaged walkways) while on the move.
- **The Obstacle:** Current reporting channels rely on slow, fragmented bureaucracy (obscure municipal web forms, phone calls) that lack real-time geographical visibility and resolution tracking. This creates a discouraging "black hole" effect for citizens.

2. Competitive Analysis & Evidence-Based Differentiation

We evaluated existing platforms across Poland and internationally:

- **Italian/International Apps (e.g., WeDU!, IOparteciPA):** Many are strictly geofenced to specific municipalities, suffer from poor UX/UI leading to low app store reviews, or lack active maintenance.
- **miasto.app & NaprawmyTo.pl (Poland):** Web-based forms that simply forward reports to city offices via email. They lack dedicated native mobile performance, real-time public map visibility, and transparent status tracking.
- **mObywatel (Poland):** Official public app, but its reporting scope is strictly restricted to environmental violations (e.g., illegal dumping), omitting daily road hazards.
- **Local Municipal Portals (e.g., lckm.uml.lodz.pl/zgl in Łódź):** Emergency web pages that are cumbersome on mobile devices and closed to public viewing.

UrbanFix Differentiation: UrbanFix provides a mobile-first, transparent public map that informs the community immediately, reduces duplicate reports through community upvoting, and maintains an up-to-date hazard map.

3. Challenging Assumptions (AI & Peer Critique)

- **Assumption 1 (Data Quality & Duplicate Control):** Users will submit accurate, constructive reports without overwhelming the system with duplicates.
    - *Critique & Mitigation:* To prevent spam and duplicate clutter, UrbanFix combines required category selection with one-tap community verification ("I saw this too"). Repeated community confirmations validate issue severity and eliminate the need for redundant pins.
- **Assumption 2 (Low-Barrier Reporting & User Privacy):** Users will report hazards quickly without being deterred by strict friction or permission prompts.
    - *Critique & Mitigation:* Mandating photos or requiring strict GPS permissions creates high reporting friction. In UrbanFix, photo uploads are completely optional, reinforced with encouraging microcopy ("Photos are optional, but help prioritize intervention"). Location input is highly flexible, supporting automatic GPS detection, direct map pinning, or manual address typing.
- **Assumption 3 (Institutional Feedback vs. Community-Driven Cleanup):** Municipalities will provide timely status updates to citizens.
    - *Critique & Mitigation:* Municipal channels rarely update resolution progress, leading to a "black hole" user experience. UrbanFix shifts from unreliable municipal lifecycle badges ("In Progress", "Resolved") to community-driven transparency. Reports display relative time tracking (e.g., "Reported 3 days ago", "Just Reported") and are governed by an automatic 30-day cleanup policy: unconfirmed reports expire automatically, ensuring the map remains accurate and fresh.

4. Minimum Viable Product (MVP) Scope

Primary User Journey

Open application → Explore interactive local map → Submit report via direct map click (Quick Report) or the dedicated Report form (with optional photo, visual category selector, and flexible location) → Confirm submission → View updated pin on the map and track real-time activity inside the Tracker feed with community upvoting.

Core Screens & Features (3 Core Screens)

1. **Interactive Map / Feed Screen:** Displays active hazard pins with category markers, user location indicator, street search bar, and direct map-tap reporting.

2. **Report Creation Screen:** Optional photo attachment, flexible location selector (Current GPS, Map selector, Text input), brief description field, and a visual category grid restricted to 4 macro-areas:
    - **a.** Roads (potholes, sidewalk damage)
    - **b.** Waste (overflowing bins, litter)
    - **c.** Lights (broken streetlights, dark zones)
    - **d.** Other (general safety hazards)
3. **Ticket Tracker / My Reports Screen:** Centralized community feed showing submitted tickets, relative time indicators, street locations, and an interactive "I saw this too" confirmation button.

Explicitly Out of Scope

- Public user-to-user commenting sections and social forums.
- Gamification mechanics, user leaderboards, and achievement badges.
- Direct municipal worker dispatch and internal route planning APIs.

5. Three User Stories & Acceptance Criteria

- **User Story 1 - Reporting Infrastructure Hazards:** As an active commuter, I want to submit a localized report with a category and optional photo/location tag, so that responsible teams and fellow citizens know the exact nature of the hazard.
    - *Acceptance Criteria:* The application allows optional photo attachments, accepts location inputs (GPS, manual text, or map click), validates required category selection, and stores the ticket upon tapping "Submit".
- **User Story 2 - Consulting the Public Hazard Map:** As a city traveler, I want to view active issues on an interactive local map, so that I can anticipate road hazards and inspect details by tapping on pins.
    - *Acceptance Criteria:* The main screen displays interactive map markers corresponding to open tickets; tapping a marker opens a detail card displaying the issue category, address, description, and community confirmation options.
- **User Story 3 - Community Verification & Tracker Overview:** As a citizen, I want to review active reports and confirm hazards I personally observed, so that issue severity is highlighted without creating duplicate reports.
    - *Acceptance Criteria:* The "Tracker" view presents submitted reports alongside relative timestamps, location details, and an interactive "I saw this too" button that updates the ticket timestamp and locks repeat votes.

6. Team Roles & Initial Task Allocation

Team Member	Role	Key Responsibilities
Antonio	UI/UX & Frontend Lead	Design and layout of Jetpack Compose screens (Map Screen, Report Form, Tracker Views), visual styling, and WCAG accessibility integration.
Diego	Application Logic & Architecture Lead	Data models, local state management, location/GPS logic, map pin positioning, and 30-day cleanup policy architecture.
Efe	Project Manager & Repository Lead	Git repository hygiene, merging branches, conflict resolution, QA testing on emulator/devices, and submission documentation.

7. Mobile UX & Accessibility Implementation

To ensure compliance with accessibility standards and provide an inclusive experience:

- **Dynamic Text Scaling (A+ Toggle):** Enables text magnification across titles, labels, input fields, and ticket cards using adaptive flex layouts to prevent text truncation or layout breakage.
- **High Contrast Mode (◐ Toggle):** Provides enhanced visual contrast for low-vision users by applying high-contrast dark color palettes across phone frames, cards, banners, and map overlays.
- **Assistive Technology & Localization Mock (EN/PL Toggle):** Integrates standardized ARIA roles (e.g., `role="tabpanel"`, `aria-label`) for screen readers, alongside an EN/PL language selector with mock feedback preparing the app for full Polish localization.

8. Prototype Peer Review & Iterative UX Enhancements (User Testing Feedback)

Based on feedback during peer testing and prototype iterations:

- **Direct Map Interaction:** Added instant "Quick Report" modal triggering upon clicking empty map coordinates.
- **Reduced Friction:** Removed mandatory photo validation in favor of clear guidance microcopy, and added multiple location input methods.
- **Duplicate Control & Data Expiration:** Replaced static municipal progress badges with community upvoting ("I saw this too") and explicit 30-day map cleanup policy transparency.